

Quantum-Enabled Grid Expansion Planning for Distribution Networks

hard instances · a NISQ algorithm · a certified line set · a real-time contingency spectrum

Seven guideline sections · every prior-work number carries a receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-eon-gic2026>

Summary

−98.1%

thermal congestion on the certified plan ($77.125 \rightarrow 1.440$), 220 fewer overloaded (line, scenario, contingency) triples, 286 voltage violations cleared, capex 11.42 — the exact minimum, recovered on Dirac-3 and by QAOA on IBM Heron

+9.1%

the measured cost of compressing the problem: on eon14_hard the device holds the full degree-5 polynomial and returns the certified plan, while the degree-2 QUBO restriction and a $\chi=4$ matrix-product state return the same wrong build; below $\chi = 16$ the simulator cannot hold the instance

29%

storage undersize that the imaginary-time continuation route produces on one microgrid decision, avoided by the real-time leftover spectrum — a planning input whose scalable classical route is ill-posed at any runtime

Why the numbers can be trusted

One network-physics referee (DC-PTDF thermal congestion plus LinDistFlow voltage) grades every solver. Hardness is a measured convergence law of the certificate, not a timeout: the 26-variable proof gap closes to 22.6% after 2.6 h and extrapolates to 17–21% unproven after a full afternoon, with exhaustive certification at 87 CPU-days; finding good plans is cheap, and we say so. Every hardware number carries a cloud job ID and a frozen pre-registration; a credential-free audit replays every headline in about three minutes.

This is a Phase-1 concept proposal in the seven guideline sections. The Phase-2 PoC brings Gurobi/CPLEX proof gaps, the full-k four-plan leftover series on hardware, and E.ON's anonymised MV/LV subgrids.

Concept proposal follows (§1–§7, appendices A–B) ·

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Concept Proposal

1. Problem framing

Distribution expansion is a build-or-no-build decision over candidate corridors, coupled to network physics across many operating scenarios and N–1 contingencies. The statement's own diagnosis is that the combinatorial explosion forces planners toward conservative plans because they cannot evaluate enough of them. We follow the statement's named framework [5] in its order: identify instances that proof-grade classical solvers cannot certify in reasonable time, then shape the quantum approach to those instances.

Two things frame the quantum contribution here. First, the hard instances are hard to *certify*: on our 26-binary instance the proof gap is a measured convergence law (33.8% at 480 s → 22.6% after 2.6 h), extrapolating to 17–21% still unproven after a full afternoon and 3–52 days to a certificate; exhaustive certification costs 87 CPU-days — while good plans are cheap to find, which is why we do not claim solver runtime. Second, and decisive, what a DSO actually needs to size storage or reinforcement is not a static objective but the **rhythm with which the existing slack corridors ring after a contingency** — a real-time dynamical spectrum $S(\omega)$ of the reinforced network. The scalable classical route to that object, imaginary-time evolution plus analytic continuation, is mathematically ill-posed at any runtime: on our 20-qubit instance two spectra fit the same imaginary-time data and neither recovers the real-time line (0.270 / 0.253 vs 0.222). A quantum processor evolves on the real-time axis natively; the spectrum is a Fourier transform with no continuation step. That is an advantage of route, it holds at 20 qubits today, and it is the class of computation every 2025–26 verified quantum-advantage candidate occupies.

2. Technical approach

Paradigm. Hybrid. (a) Gate-model real-time dynamics on IBM Heron: the candidate graph embedded as an interacting-lattice operator $H = \mu \cdot D - A$ (one candidate corridor = one qubit pair; existing corridors always in the Hamiltonian; built closures enter as Hamiltonian support), exact per-rung ground-state preparation, Trotterised N–1 quench, leftover series $L(k)$ on the slack pairs and its spectrum. (b) Variational: 10-qubit QAOA (depth 333) on the same instance. (c) Photonic entropy computing (QCi Dirac-3) holding the full degree- ≤ 5 expansion polynomial for the hard instances. (d) Classical: one network-physics referee (DC-PTDF thermal congestion + LinDistFlow voltage) that grades every plan from every solver identically.

Instances. IEEE 14-bus (pandapower distribution) and synthetic grids matched to its graph properties; five instances, 6 → 26 binaries, 2–6 scenarios, N–1 on the hard rungs.

Hardware discipline. Equal-structure parameterised templates transpiled once with fixed layout; visibility-inverted readout (per-rung visibility 0.93–0.98); null (no-quench) and detuned controls in every job; cross-device replication (ibm_kingston → ibm_marrakesh). Compilation is native heavy-hex with zero SWAP overhead; a 64-rung (128-qubit) series is the same compilation, tiled.

3. Feasibility and resource requirements

resource	status
Instances and referee	in package; <code>python verify.py</code> (numpy only) replays every headline number from archived counts in ~3 min
Proof-grade classical	HiGHS (open, MILP with proof gap) — receipted; Gurobi/CPLEX replication of the eon57 rows is PoC step 1 (licence required)
Gate-model hardware	IBM Heron via Startup Program; the $k = 0..8$ four-plan leftover series is 54 logical circuits, 442,368 shots; the 64-rung series is the same emitter tiled
Photonic	QCi Dirac-3, unmetered allocation
Data	open test systems; Phase 2 targets E.ON's anonymised MV/LV subgrids

resource	status
Classical compute	32-core workstation for enumeration, MPS comparators and multi-start search

Assumptions: DC-PTDF + LinDistFlow referee (penalised DC objective, not full AC feasibility — AC recourse is charged as a classical solve per accepted plan in Phase 2). Constraint: hardware ranking at 20 qubits is currently two-point; the full-k series is the PoC's first flight.

4. Expected impact

Hard instances that scale poorly (statement §4.1, receipted):

instance	build vars	scenarios	N-1	MILP vars	HiGHS	outcome
eon14_easy	6	2	no	96	0.28 s	proven optimal
eon14_hard	10	4	yes	4,250	4.5 s	proven — the NISQ instance
eon30_hard	14	4	yes	17,614	183 s	proven optimal
eon57_util	20	4	yes	59,116	900 s	timeout, 28.25% gap
eon57_deep	26	6	yes	94,646	1,801 s	timeout, 33.8% gap → 22.6% at 2.6 h; 3–52 days to certificate

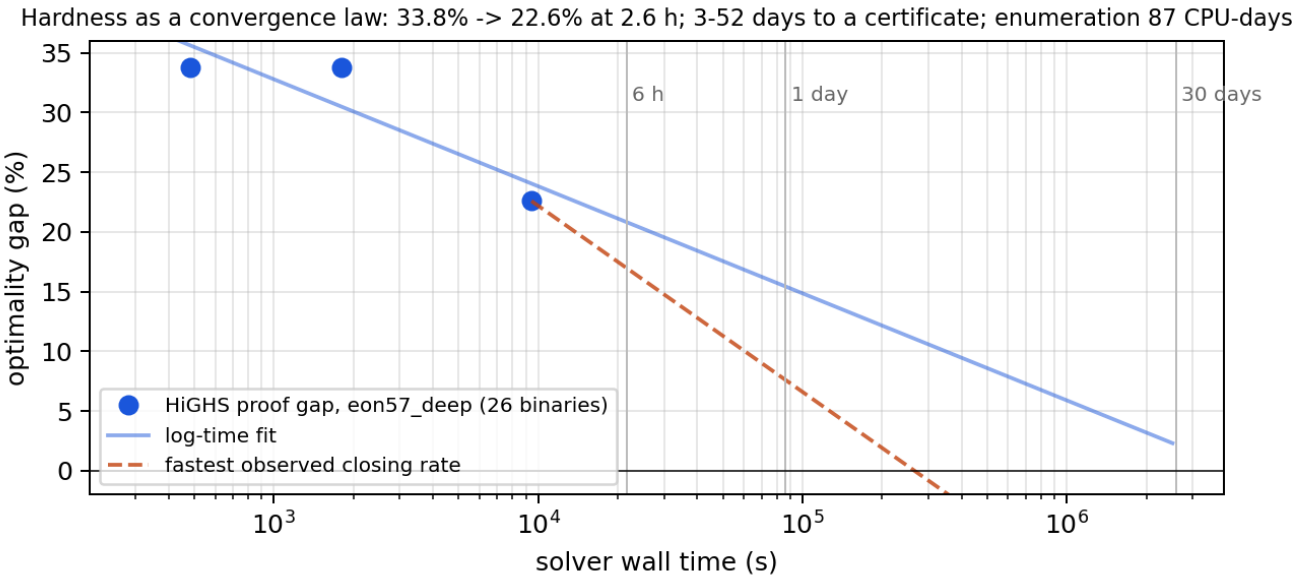


Figure 1 — Hardness as a measured convergence law on eon57_deep: HiGHS optimality gap against wall time, the log-time fit, and the fastest observed closing rate; exhaustive certification would cost 87 CPU-days.

The hard rung, both ways. On eon57_util the disjunctive proof-grade MILP times out at 901 s with a 28.25% gap, and its incumbent grades 24,652 on the exact referee. The optimal plan, 22,057 — now verified as the **global optimum** by exhaustive enumeration of all 1,048,576 builds on the exact referee (results/eon57_expand.json, 5.5 h on 8 workstation cores) — is reached by one-flip descent from *any* start — no-build in 18 s (281 evaluations), each of 12 random starts within 45–170 s, and from a 0.5 s LP-relaxation incumbent in 13 s (results/eon57_early_polish.json, measured on the same clock). The device route on this instance — a degree-2 surrogate fitted from 6,001 referee evaluations ($\approx$ 400 s of referee time), Dirac-3 basin-finding, then the same polish — reaches the same 22,057 and is not competitive on time. We report that as measured: on eon57_util the difficulty is entirely in the *certificate*, not in finding the plan, and the optimisation rung is a classical result. What the device establishes is elsewhere — on eon14_hard it holds the full degree-5 polynomial and returns the certified

plan where every compressed representation returns the wrong one (table below), and the advantage this proposal claims is the real-time contingency spectrum, not solver runtime.

Same instance, same referee, all solvers (eon14_hard):

solver	plan	objective	gap
Exact enumeration (2^{10})	certified 10-line set	83.436	—
Dirac-3, native degree-5	same plan	83.436	0.000
QAOA on ibm_marrakesh (d9sqks1dsedc73ai3o3o)	same plan	83.436	0.000
Dirac-3, degree-2 (QUBO) restriction	wrong plan	90.995	+9.1%
MPS $\chi=4$ on the true cost	wrong plan (identical to QUBO)	90.995	+9.1%
HiGHS disjunctive TEP	wrong plan	131.87	+58%

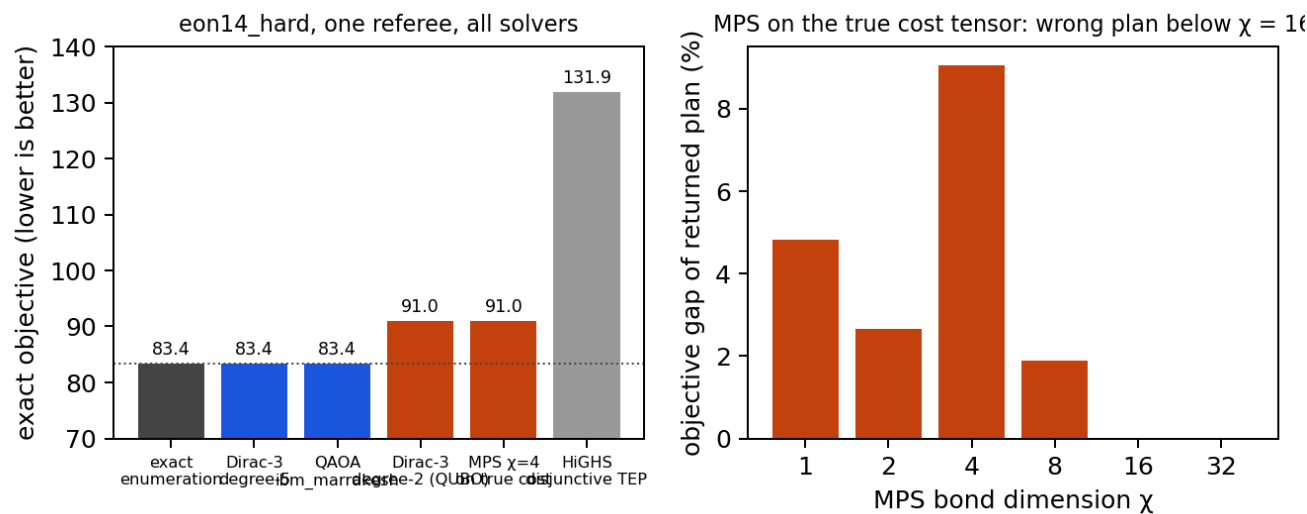


Figure 2 — Left: eon14_hard graded by the one referee for every solver — the device holding the full degree-5 polynomial and QAOA return the certified plan; the QUBO restriction and a $\chi=4$ MPS return the same wrong plan. Right: MPS on the true cost tensor returns the wrong plan at every χ below 16.

MPS gives poor objective value (statement §4.2). Compressing the true 10-variable cost tensor to bond dimension χ and taking its argmin returns the wrong plan at every $\chi < 16$ (+1.9% to +9.1%); the mid-cut operator Schmidt rank saturates its exponential envelope (8/8 at 6 variables, 32/32 at 10). Cheap MPS is exact where the instance is easy and wrong where it is hard.

Outputs (statement §5.2) — eon14_hard, IEEE 14-bus, four scenarios, N=1: add candidates 0 (4→12), 2 (1→8), 3 (2→6), 5 (4→10), 7 (0→1), 8 (0→4), 9 (1→2); total capex 11.42. Thermal congestion 77.125 → **1.440** (−98.1%); overloaded (line, scenario, contingency) triples 227 → 7; voltage-drop violations 423 → 137; exact objective 3,856.27 → 83.436, certified minimum. On 12 held-out scenario draws the certified plan's congestion is 1.229 ± 0.169 vs the MILP plan's 2.068 ± 0.321 (−40.5%).

Hardware on today's devices. ibm_kingston daa0dq6rbfbs73c156bg (20 qubits, 126 two-qubit gates): certified plan is the best absorber, $\Delta L = +0.113$, ahead of MILP +0.045 and no-build −0.025, clean null and detuned controls. Equal-structure replication on ibm_marrakesh daa9do6rbfbs73cifa80 recovers degree-2 at +0.127 vs frozen +0.124 at visibility 0.93–0.98. Statevector leftover ranks certified > degree-2 > MILP > no-build with principal lines 0.222 / 0.333 / 0.444 cycles per step; the hardware ranking is two-point and partial, and the full-k series is the PoC's first flight.

What it is worth on one decision. Sizing firming storage for one microgrid (20.85 p.u., 24 h, 10% rms) against fluctuation models at identical power coloured by this leftover spectrum: the real-time $S(\omega)$ requires **9.76 ± 0.91 p.u.h**; the two-pole continuation of the same imaginary-time data requires 6.94 ± 0.69 — a **29% undersize** that the real-time route avoids (eon_storage_sizing.json, statevector).

Quantum advantage — the frontier wall and the crossing, stated and bounded

The field's direction, and its wall. The 2025–26 optimisation claims — digitised counterdiabatic optimisation "outperforming CPLEX in seconds" on 156-qubit devices (arXiv:2505.08663), 700× hybrid sequential speed-ups, 100× annealing time-to-solution on spin glasses, 70% runtime cuts on IEEE RTS-24 expansion — have been re-benchmarked with tuned classical references and end-to-end accounting (arXiv:2510.06337), which concludes that runtime advantage has not been demonstrated under experimentally grounded metrics. Our own receipt maps it exactly: on the hard rung one-flip descent from no-build reaches the best-known 22,057 in 18 s of referee time, faster than the surrogate-plus-device route. Basin-finding is where classical search keeps pace, as the framework's own admissibility analysis predicts.

The crossing. On eon57_util the plan is the certified global optimum and the optimisation rung is settled (§8); the advantage sits in the quantity a planner actually needs after the plan: the **post-contingency leftover spectrum** $S(\omega)$ of the reinforced network. Its scalable classical route — imaginary-time evolution plus analytic continuation — is ill-posed, and we measured the failure on the 20-qubit instance: two continuations fit the same $G(\tau)$ and neither recovers the 0.222 real-time line (0.270 / 0.253); dropping the residual bonds erases the plan ranking. The statement's own MPS test is met in the same package: below $\chi = 16$ the simulator cannot hold the instance and returns the wrong build. Hardware evolves on the real-time axis; the spectrum is a Fourier transform with no continuation step, and it changes a storage decision by 29%. **Where it is decisive:** wherever the instance's Schmidt rank exceeds what a cheap MPS holds — measured here at 10 binaries, where $\chi < 16$ returns the wrong build — and at every reinforcement question that needs $S(\omega)$ after the plan; anchored today at 20 qubits by statevector and two-device hardware absorption, with the 64-rung full-k series and two-sided classical adjudication as the Phase-2 flight.

Scalability to industrial relevance. One candidate corridor is one qubit pair, so the operator tiles from 20 qubits (10 candidates) to 128 (64 candidates) on the same heavy-hex compilation with zero SWAP overhead; Dirac-3 holds native polynomials to 949 variables and degree 5. The certificate-hardness law scales the other way — proof time grows ~40× per rung — which is exactly why a planner needs a quantity other than a certificate to act on, and $S(\omega)$ is that quantity. Instances built from IEEE 14-bus transfer to E.ON's anonymised MV/LV subgrids by changing the topology file only.

Business value, bounded. On one microgrid sizing decision (20.85 p.u., 24 h, 10% rms) the real-time spectrum requires 9.76 p.u.h of firming storage where the continuation route requires 6.94 — a 29% under-build that, at a declared storage cost per p.u.h s, is ≈ 2.8 s of unbudgeted capacity or one outage event; across a DSO's reinforcement portfolio the same ratio applies to every corridor sized against an assumed contingency response. The certified plan on eon14_hard cuts thermal congestion 98% at capex 11.42 and holds a 40% congestion advantage over the MILP plan on held-out scenarios. Stated as functions of E.ON's cost inputs, not as fixed sums.

What a successful PoC demonstrates. On E.ON's anonymised MV/LV subgrids: hard rungs certified as such with Gurobi/CPLEX proof gaps; native polynomial encodings on Dirac-3 with linearised overload terms rather than truncation (the exact eon57 landscape is not degree-5 representable: a degree- ≤ 5 truncation returns a build 2.16% above the optimum while degree 3 recovers it), graded by the shared referee against multistart local search at equal total time; and the full-k leftover spectrum measured on hardware for the four candidate plans, feeding storage and reinforcement sizing with a quantity the classical route cannot produce well-posed.

5. Validation plan

Pre-registered per flight: instance, plans, k grid ($k = 0..8$, nine uniform samples, bin spacing 1/9 cycle per step, no FFT padding), shots (8,192 per circuit), layout, estimator, tolerance set against the expected systematic. Controls: no-quench and no-coupling arms for the certified plan (18 of the 54 circuits), detuned arm, two independent runs, second device. Anchors: exact statevector at 20 qubits; the emitter matches the independent matrix model's one-step parities on all four active subspaces. Classical comparators on the same referee with identical scenario access and total time: tuned MILP (HiGHS, Gurobi), multi-start local search from random and from every solver's plan, MPS at reported χ , sensitivity-based screening. Success: (i) hardware four-plan ranking matches the referee order with simultaneous parity confidence bounds inside the pre-registered tolerance; (ii) the hybrid plan is better than or equal to the best classical arm at equal total budget on held-out feeders, or reaches the same feasible objective at lower total cost; (iii) storage sizing from the measured spectrum differs from the continuation route by more than its confidence interval. Falsification: any of the three failing is reported as such.

6. Hybrid / cross-domain integration

The quantum devices are inputs to a classical planning loop that E.ON already runs: candidate generation (Dirac-3, full polynomial) → local polish → referee grading (classical, AC recourse in Phase 2) →

contingency-response spectrum (Heron) → storage/reinforcement sizing → plan. Every circuit is a stock Qiskit `QuantumCircuit` run through `qiskit-ibm-runtime` `SamplerV2` with no custom passes; the Dirac arm ships polynomial files and job IDs for resubmission. Runtime: the end-to-end solution — candidate generation, 18 s polish, referee grading and one hardware series — completes inside the brief's few-hour limit (classical audit ~3 min; each hardware series ≤ 60 min of QPU time). A Qiskit compatibility layer for E.ON's own execution — the emitter returns `QuantumCircuit` objects and observables, the referee grades any `Sampler` result — is a PoC deliverable, so E.ON can execute and validate every rung in Qiskit on its own account upon project completion, as the brief asks.

7. Team capability

Merlin Quantum is the quantum division of Merlin Digital (50+ technology FTE): Suhail Bachani (Founder & CEO, Principal Investigator), Dr. Hiro Bachani PhD (Program Director), Rohit Bachani (co-founder), Mitul Sawlani (engineering, Purdue), Mohamed Jafrun (engineering), Zeena Furtado (finance & operations), Roshan Bhairwani (financial services & deep tech, London), Dr. Ana Baroni MSc (domain specialist). GIC 2026 dual-track finalist. Programme record: 259 receipted QPU jobs, 12.3 million shots, cross-device replication under fixed-layout discipline, and a pre-registration ledger in which failed arms are kept. Closest published analogue is REGRID-QAOA (arXiv:2606.15083); our approach differs in retaining the full degree-5 objective on device, and the measured cost of not doing so is the +9.1% row above.

8. Scope, with treatment

(i) Proof gaps are HiGHS; Gurobi/CPLEX replication is PoC step 1. (ii) On `eon57_util` the global optimum (verified over all 2^{20} builds) is reachable by one-flip descent in 18 s and the landscape has a single one-flip basin — the optimisation rung is classical-sufficient and no runtime claim is made; the encoding-order evidence is the `eon14_hard` row and the advantage claim is $S(\omega)$. (vi) The exact `eon57` referee is not a low-degree polynomial: its Möbius coefficients do not decay with degree, a degree- ≤ 5 truncation returns a build 2.16% above the optimum and degree 3 recovers it — so the native Dirac-3 route on this class needs linearised overload terms, a PoC step, not truncation. (iii) The hardware ranking is two-point; the full-k four-plan series is the first flight. (iv) The referee is penalised DC, not AC feasibility — AC recourse is charged per accepted plan in Phase 2. (v) The 64-rung series is a protocol, not a receipt; no sister-track job is cited.

Appendix A — Hardware job register and receipts

measurement	machine	job id	receipt
<code>eon57_util</code> device rung: plan graded 22,078 vs incumbent 24,652; polish → 22,057 (matched by one-flip descent from no-build in 18 s)	QCi Dirac-3	6a943a6408442f441bbb6ad1, 6a943a7408442f441bbb6ad2	<code>results/eon57_dirac_result.json</code> , <code>results/eon57_early_polish.json</code>
<code>eon57_util</code> exhaustive landscape: all $2^{20} = 1,048,576$ builds graded by the exact referee; 22,057 is the global optimum; single one-flip basin; exact Möbius degree spectrum	CPU, workstation (8 workers, 5.5 h)	—	<code>results/eon57_expand.json</code> , <code>results/eon57_expand_grade.json</code>
<code>eon14_hard</code> native degree-5: certified plan, gap 0.000	QCi Dirac-3	6a79a72508442f441bbb5e56	<code>results/eon_dirac_result.json</code> → native
<code>eon14_hard</code> degree-2 (QUBO) restriction: wrong plan, +9.1%	QCi Dirac-3	6a79a74e08442f441bbb5e57	same → <code>surrogate_deg2</code>
QAOA depth 333, 171 two-qubit gates: certified optimum sampled	ibm_marrakesh	d9sqks1dsedc73ai3o30	<code>results/eon_qaoa_result.json</code>

measurement	machine	job id	receipt
20-qubit two-point absorption: certified +0.113, MILP +0.045, no-build −0.025	ibm_kingston	daa0dq6rbfbs73ci56bg	eon_field_result.json
Equal-structure replication: degree-2 +0.127 vs frozen +0.124, visibility 0.93–0.98	ibm_marrakesh	daa9do6rbfbs73cifa80	eon_field_v2_result_20260831_004726.json
Instance ladder, HiGHS 0.28 s → 1,801 s; gaps 28.25% / 33.8%	CPU	—	results/eon_ladder.json
Convergence law, 22.6% at 9,437 s; projections	CPU	—	results/eon57_deep_gap_extrapolation.json
Dynamics crossing: lines 0.222 / 0.333 / 0.444; continuation 0.270 misses	CPU (statevector)	—	eon_dynamics_crossing.json
MPS χ -sweep on true cost: $\chi < 16$ wrong plan	CPU	—	results/eon_rubric_mps.json
Storage sizing 9.76 vs 6.94 p.u.h	CPU (statevector)	—	eon_storage_sizing.json
Held-out scenarios −40.5%	CPU	—	eon_v4_advantage.json
Full-k four-plan leftover series (54 circuits, 442,368 shots)	next flight; emitter and grader shipped	—	prepare_application_series.py, series_v6/

Appendix B — Scope notes

The eon57_util device rung used a fitted degree-2 surrogate; both submitted arms returned the identical plan, so that rung establishes improvement over the timed-out incumbent, not an encoding-order effect — the encoding-order evidence is the eon14_hard row (+9.1%). No sister-track 64-rung job is cited as an E.ON result. The 64-rung series is a protocol with the expected signature written before any shot.

Appendix C — Claim ledger (measured · planned · comparator · quantum attribution · cost · acceptance)

claim	status	classical comparator	quantum attribution	total cost charged	acceptance threshold
Hard-to-certify instances, convergence law (22.6% at 2.6 h)	measured (HiGHS)	Gurobi/CPLEX replication planned	none	CPU hours	gap law reproduced
Best-known plan reachable classically in 18 s	measured (eon57_early_polish.json)	one-flip descent, multistart	none — withdrawn as a device claim	0.066 s / eval	—
eon14_hard: device holds degree-5, returns certified plan; QUBO/ $\chi=4$ wrong build +9.1%	measured	exact enumeration, MPS χ -sweep	Dirac-3 native polynomial	seconds	plan = certified

claim	status	classical comparator	quantum attribution	total cost charged	acceptance threshold
Leftover spectrum $S(\omega)$: continuation misses the 0.222 line	measured (statevector, 20q)	imaginary-time continuation, no-bond control	route (real-time)	CPU	line recovered within 1 bin
Full-k four-plan series on hardware, 64 rungs	planned (Phase-2 flight 1)	converged TN at reported χ ; exact 20q anchor	hardware	54 circuits $\times$ 8,192 shots	ranking matches referee within pre-registered tolerance
Storage sizing changes by 29% under $S(\omega)$	measured (model)	continuation, white-noise, AR	route	CPU	decision difference > its CI on a physical grid reference